

Student Performance Analytics Dashboard - Insights Report

Project Overview

This project performs Exploratory Data Analysis (EDA) on a student performance dataset and presents the results through an interactive Streamlit dashboard.

Dataset Description

The dataset contains student demographics, study habits, attendance records, career aspirations, and subject scores. An average score feature was created to evaluate overall academic performance.

Data Cleaning Process

- Handled missing values
- Removed duplicate records
- Created average_score feature
- Detected and removed outliers using the IQR method

Key Findings

1. Students with higher self-study hours generally achieve better academic performance.
2. Higher absence days tend to reduce overall scores.
3. Subject scores are positively correlated.
4. Career aspirations vary significantly across students.
5. Part-time jobs have a limited impact on academic performance.

Conclusion

Study habits and attendance are important factors influencing student performance. The dashboard provides an interactive way to explore these patterns and supports data-driven educational insights.